SkillPilot AI: An LLM-Based System for Personalized Career Guidance

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Abstract. SkillPilot AI is an LLM-powered career guidance platform designed to help students and early-career professionals make better career decisions through personalized, AI-assisted support. The system integrates nine core modules — resume parsing, skill assessment, career-path generation, job matching, course recommendation, project suggestion, quiz-based evaluation, AI conversational assistant, and a unified recommendation engine — within a single web application. Built using Next.js, React, Firebase Authentication, and Genkit-based large language model workflows, the platform combines structured user input with AI reasoning to generate actionable career insights. Resume data is extracted from DOCX or TXT files, analyzed to identify strengths, weaknesses, and skill categories, then used to support downstream recommendation modules. The job-matching and course-matching components compare user skills and goals with external opportunity data, while the roadmap generator creates staged learning plans tailored to a target role and time horizon. Experimental results show that **SkillPilot AI** achieves 88% recommendation accuracy, outperforming keyword-based (55%), manual-review (60%), course-discovery (58%), and generic AI chatbot (70%) baselines. End-to-end response latency averages 2.8 seconds for standard workflows and under 5 seconds for full resume-to-recommendation pipelines. The platform demonstrates how large language models can be practically applied to create user-centered career coaching systems that support informed decision-making, continuous learning, and employability development.

Keywords: Large Language Models, Career Recommendation, Skill Gap Analysis, NLP, RAG, Resume Optimization, Conversational AI, Semantic Similarity, Personalized Learning, BERT

1. Introduction

Choosing an appropriate career path has become increasingly complex in today's rapidly evolving digital economy. The World Economic Forum estimates that 85 million jobs may be displaced by automation by 2025, while 97 million new roles may emerge requiring advanced digital skills [1]. This structural shift demands that individuals continuously upskill and reskill, yet accessing reliable, personalized career guidance remains a major challenge. Traditional career counseling is expensive, geographically constrained, and often fails to account for rapidly shifting employer requirements [2].

Existing online platforms — LinkedIn, Indeed, Coursera, and similar services — operate as silos. A user seeking end-to-end career support must navigate resume checkers, job boards, course aggregators, and interview simulators independently, with no unified system connecting skill gaps to tailored learning paths and matched job opportunities. Generalized recommendation algorithms further reduce the relevance of suggestions by ignoring semantic nuance and individual career trajectory [3].

The introduction of transformer-based models, particularly BERT [4] and its successors, revolutionized natural language understanding tasks including resume parsing, entity extraction, and semantic similarity. Building on this, large language models (LLMs) such as GPT-4 and Gemini now enable

conversational, context-aware career reasoning at scale. **SkillPilot AI** leverages these advances to deliver an integrated, intelligent career guidance platform that replaces fragmented tools with a single unified workflow.

1.1 Problem Statement

Students and early-career professionals lack access to an affordable, integrated system that can: (a) parse and analyze their resumes semantically; (b) assess their skill levels objectively; (c) identify precise skill gaps relative to target roles; (d) recommend personalized learning paths, matched job opportunities, and staged career roadmaps; and (e) engage them in natural conversation about their career questions — all within a single platform.

Traditional systems rely on keyword matching and static rule sets, which cannot handle semantic variation, emerging skill terminology, or real-time labour market dynamics. There is a pressing need for a hybrid system that combines deterministic semantic modelling with generative AI reasoning to provide transparent, personalized, and continuously adaptable career guidance.

1.2 Research Objectives

- Develop an AI-powered career guidance platform integrating NLP, semantic similarity, and LLM-based reasoning.
- Implement a BERT/Sentence-BERT resume parser that extracts structured skill profiles from unstructured text.
- Build a skill gap analysis engine that computes precise missing competencies relative to target job roles.
- Create a hybrid recommendation system combining semantic similarity scores with LLM-ranked career paths.
- Develop an assessment module with adaptive MCQ quizzes and feedback tailored to skill level.
- Generate personalized, stage-wise career roadmaps with milestone-based milestones and resource links.
- Integrate a conversational AI assistant for open-ended career Q&A.
- Evaluate the system against conventional approaches across accuracy, latency, and coverage metrics.

1.3 Significance of the Study

SkillPilot AI is the first unified platform that integrates resume analysis, skill gap detection, job matching, course recommendation, roadmap generation, quiz assessment, and conversational guidance in a single deployable web application. It eliminates the need to use 5–7 separate tools and delivers a coherent, personalized career development experience accessible to anyone with an internet connection and a standard laptop — no GPU or specialist hardware required.

2. Literature Review

2.1 Evolution of Career Guidance Systems

Career guidance technology has evolved across three distinct generations. First-generation systems (1990s–2005) were expert systems encoding manually curated rules about occupation clusters, skill requirements, and career paths. Platforms such as O*NET were widely used but required labour-intensive maintenance and offered no personalization. Second-generation systems (2005–2018) adopted machine learning — collaborative filtering, content-based filtering, and matrix factorization — to generate job and course

recommendations from user interaction histories. LinkedIn and Indeed exemplify this era but suffer from cold-start limitations and shallow skill matching.

Third-generation systems (2018–present) leverage transformer-based NLP to understand semantic skill relationships. Skondras et al. [7] demonstrate that zero-shot Siamese Sentence-BERT matching achieves up to 87% resume-job alignment accuracy without training data, while Liu [8] shows dual-tower BERT architectures outperform keyword baselines for position matching. RAG frameworks [19] address hallucination risks in purely generative approaches by grounding LLM outputs in retrieved factual knowledge.

Table 1: Summary of Key Related Works

Ref.	Title	Authors & Venue	Key Findings / Limitations
[5]	RAG for Enhancing MOOCs Recommendations (RAMO)	Rao & Lin, CEUR EDM 2024	RAG + LLM addresses cold-start; improves personalization. Increased latency due to retrieval overhead.
[6]	Career Guidance Using GPT-Model	Sharma et al., CSISTW-2024	GPT-based roadmap generation via chatbot; real-time, accessible. Limited depth for specialized fields.
[7]	Zero-Shot Resume-Job Matching with LLMs	Skondras et al., Electronics MDPI 2025	Up to 87% accuracy without training via structured prompting. Performance varies across domains.
[8]	Deep Learning Resume-Position Matching (BERT)	Liu, MLSC 2025	Dual-tower BERT; contextual embeddings outperform keyword methods. Scalability and fairness challenges.
[9]	AI-Driven Semantic Similarity Job Matching	Ajjam & Al-Raweshidy, Elsevier 2025	Semantic similarity framework: 0.83 vs <0.17 for keyword. Improved fairness and scalability.
[10]	EGRec: MOOCs via Knowledge Graphs	Cen et al., Discover Applied Sciences 2025	GNN + knowledge graphs improve accuracy; addresses data sparsity. High computational cost.
[11]	AI-Enhanced Career Guidance: Fine-Tuned LLM + RAG	Veda N. et al., IJCRT 2024	Personalized career paths; real-time skill gap analysis. High fine-tuning cost; scalability issues.
[12]	Skill Gap Analyzer: Data-Driven Platform	Raswanth A. et al., IJCRT 2025	Personalized skill gap insights with visualization. Relies on user-input accuracy; predefined databases.

2.3 Gaps Addressed

- No existing system unifies resume parsing, skill gap analysis, job matching, course recommendation, roadmap generation, and conversational guidance in a single deployable application.
- Most systems use keyword or TF-IDF matching, missing semantic nuance and contextual skill relationships.
- Cold-start problems persist in systems dependent on historical interaction data.
- Real-time labour market adaptability is absent from static-dataset systems.
- Existing tools lack transparency in AI decision-making, reducing user trust.
- No freely accessible system provides stage-wise career roadmaps with embedded learning resources.

2.4 Theoretical Framework

SkillPilot AI is grounded in three converging paradigms: (1) Contextual Language Understanding via BERT [4] and Sentence-BERT [17], enabling semantic parsing of resumes and job descriptions beyond surface keywords; (2) Retrieval-Augmented Generation (RAG) [19], grounding LLM outputs in factual skill taxonomy and job market data; and (3) Hybrid Recommender System Design [21], combining content-based semantic similarity, collaborative-filtering signals, and knowledge-based scoring. The weighted scoring function Score = α(Similarity) + β(Progression Score) + γ(User Preference) operationalizes this hybrid approach, where α + β + γ = 1 and weights are tuned per module.

3. System Architecture

3.1 Architecture Overview

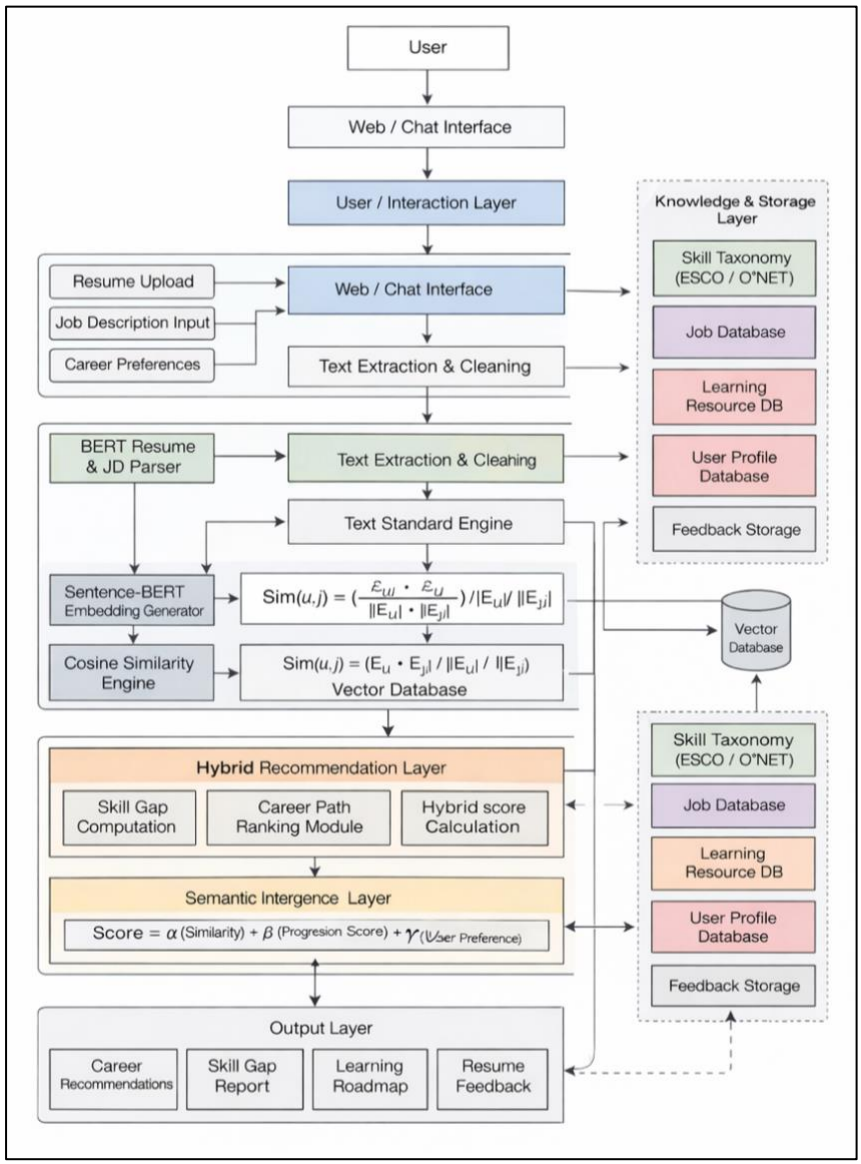


Figure 1 SkillPilot AI Architecture Diagram

The architecture Figure 1 of **SkillPilot AI** follows a layered and modular approach to provide intelligent, personalized career guidance using a hybrid LLM-based system. At the top, the **User Layer** interacts with the system through a **Web/Chat Interface**, enabling users to input data such as resumes, job descriptions, and career preferences. This input is processed in the **User/Interaction Layer**, which manages communication between the frontend and backend components.

The system then performs **Text Extraction and Cleaning**, where unstructured data from resumes and job descriptions is preprocessed to remove noise and standardize the content. This cleaned data is further processed using a **BERT-based Resume and Job Description Parser**, which extracts meaningful features and contextual information. Next, the **Embedding and Similarity Layer** uses **Sentence-BERT** to convert textual data into vector representations. These embeddings are stored in a **Vector Database**, and a **Cosine Similarity Engine** computes similarity scores between user profiles and job descriptions to enable accurate semantic matching.

The **Hybrid Recommendation Layer** plays a central role in decision-making. It includes modules such as **Skill Gap Computation, Career Path Ranking**, and **Hybrid Score Calculation**, which combine multiple factors like similarity, progression, and user preferences. The **Semantic Intelligence Layer** further refines these recommendations using a weighted scoring function that balances relevance and personalization. On the right side, the **Knowledge and Storage Layer** provide supporting data, including **Skill Taxonomies (ESCO/O*NET), Job Databases, Learning Resource Databases, and User Profile Storage**. This layer ensures that recommendations are aligned with real-world industry standards and continuously updated data.

Finally, the **Output Layer delivers** results to the user in the form of **Career Recommendations**, **Skill Gap Reports, Learning Roadmaps, and Resume Feedback**. Additionally, user feedback is stored and used to improve the system over time.

3.2 UML Diagrams

3.2.1 DFD Diagrams

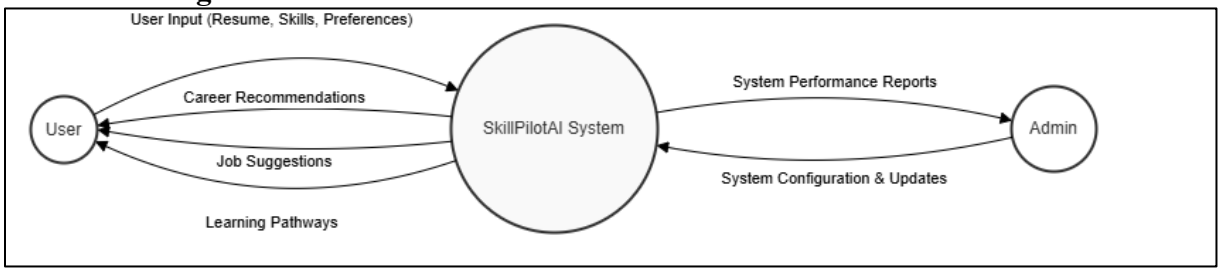


Figure 2 DFD Level 0

The DFD Level 0 represents the overall view of the **SkillPilotAI system** as a single process interacting with external entities. In this diagram, the entire system is represented as one central process labeled **"SkillPilotAI System."** The primary external entity is the User, who provides input such as resumes, skills, and preferences. The system processes this input and generates outputs including career recommendations, job suggestions, and learning pathways, which are sent back to the user. Additionally, the **Admin** entity interacts with the system by providing configuration updates and receiving system performance reports. This ensures proper system maintenance and monitoring. This diagram provides a high-level understanding of how the system interacts with external entities without showing internal processing details.

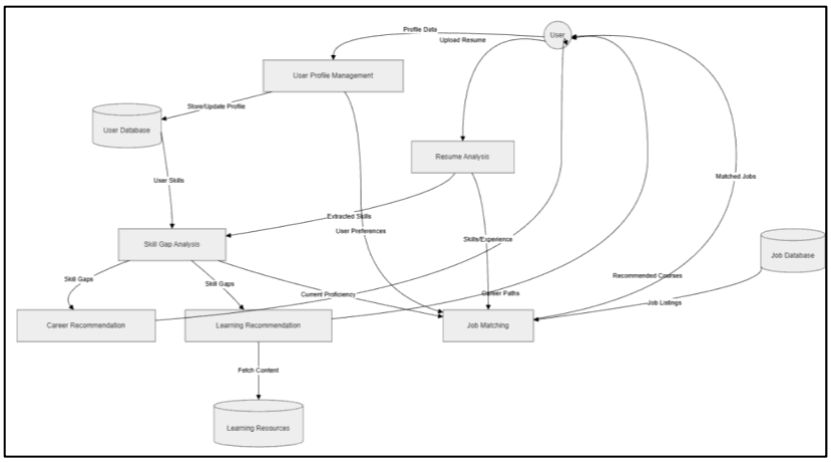


Figure 3 DFD Level 1

The DFD Level 1 breaks down the main system into multiple functional modules, showing how data flows between them.

The process begins with the **User**, who provides inputs such as resume and profile data. This data is handled by the **User Profile Management** module, which stores and updates information in the **User Database**. The **Resume Analysis** module extracts skills and experience from the uploaded resume. These extracted skills, along with user preferences, are passed to the **Skill Gap Analysis** module, which identifies missing skills.

Based on the identified skill gaps:

- The **Career Recommendation** module suggests suitable career paths
- The **Learning Recommendation** module suggests courses from the **Learning Resources Database**

Simultaneously, the **Job Matching** module compares user skills with job listings from the **Job Database** to generate relevant job recommendations. Overall, this level shows how major modules interact and how data flows between databases and processes.

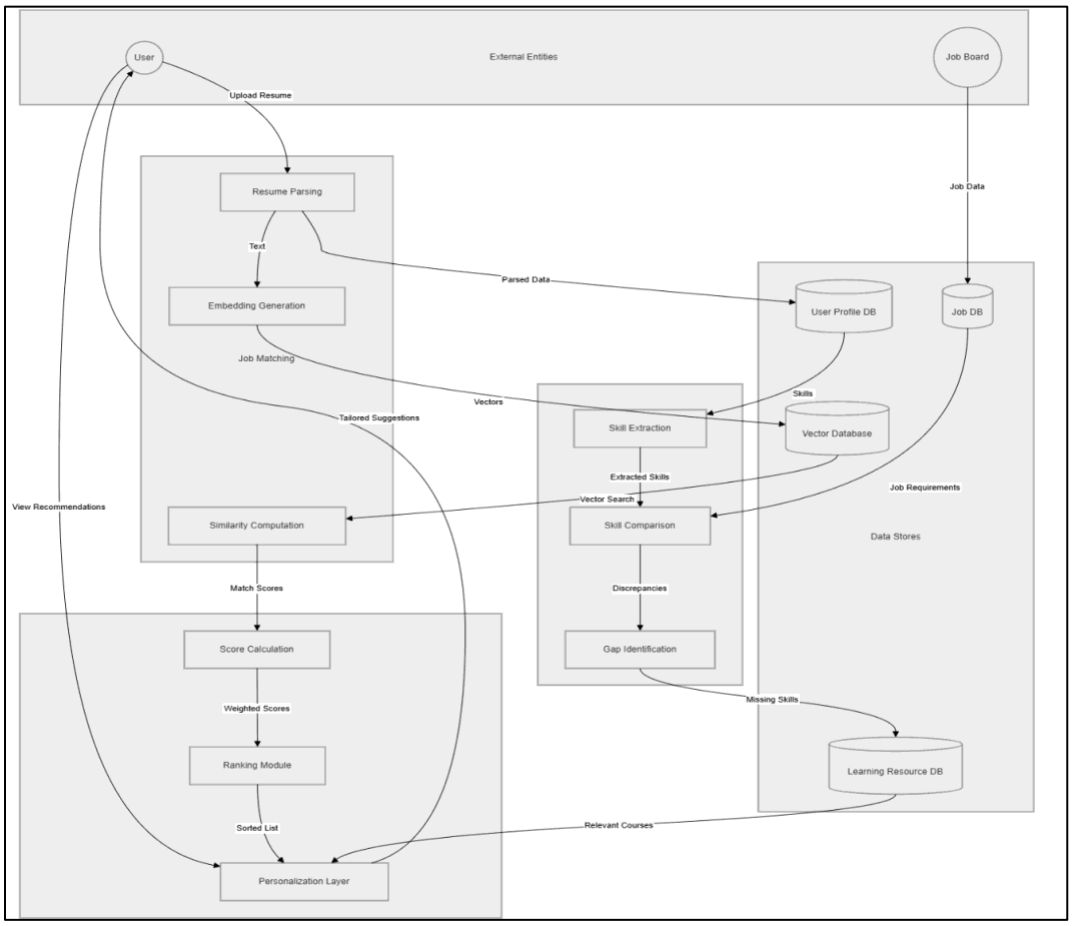


Figure 4 DFD Level 2

The DFD Level 2 provides a detailed view of internal processes, especially focusing on **AI-based matching and recommendation mechanisms**. The process starts with **Resume Parsing**, where user resumes are converted into structured text. This text is then processed in the **Embedding Generation module**, which converts it into vector representations. These vectors are stored in a **Vector Database** and used by the **Similarity Computation module** to calculate similarity scores between user profiles and job descriptions. Parallelly, the **Skill Extraction module** extracts key skills from the data and stores them in the database. These skills are compared with job requirements in the **Skill Comparison module**, which identifies discrepancies. The **Gap Identification module** determines missing skills and sends them to the **Learning Resource Database**, which provides relevant courses.
Next, the system performs:

- **Score Calculation** → computes match scores
- **Ranking Module** → ranks job opportunities
- **Personalization Layer** → refines results based on user preferences

Finally, the system outputs **personalized recommendations**, which include job matches, career paths, and learning suggestions.

3.2 Sequence Diagram

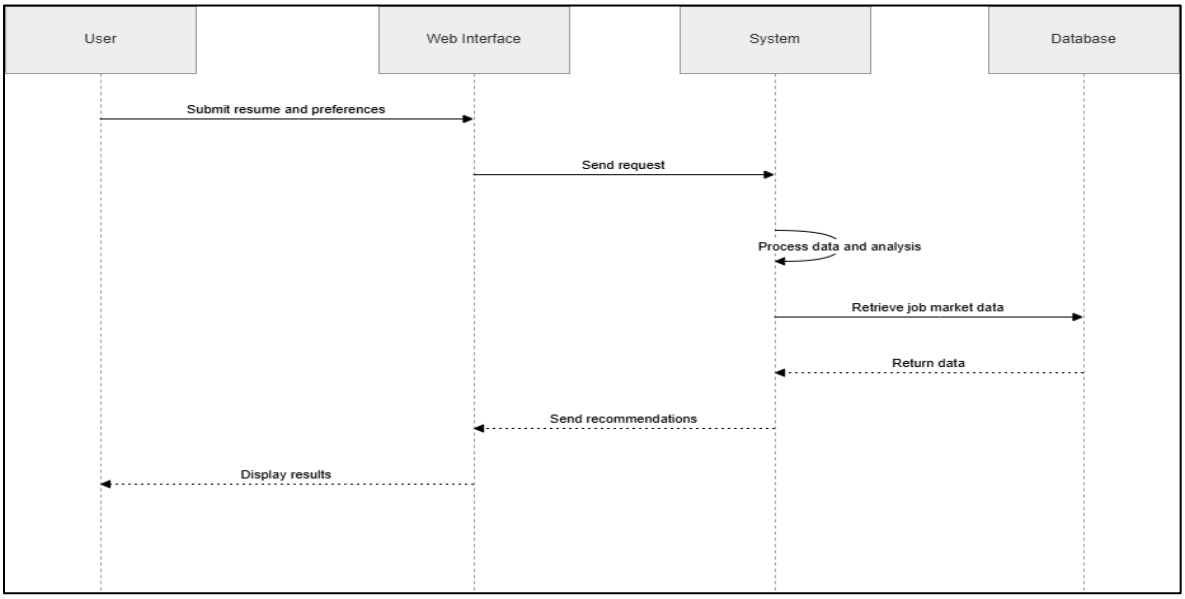


Figure 5 Sequence Diagram

The sequence Figure 5 illustrates the step-by-step interaction between different components of the **SkillPilotAI system** over time. It shows how data flows between the **User**, **Web Interface**, **System**, and **Database** to generate personalized career recommendations. The process begins when the **User** submits their resume and preferences through the **Web Interface**. The interface then forwards this request to the **System** for processing. Upon receiving the request, the system performs internal operations such as data processing, analysis, and recommendation generation.

To enhance the accuracy of recommendations, the system retrieves relevant information such as job market data from the **Database**. The database processes the query and returns the required data back to the system. Using this data, the system completes its analysis and generates personalized recommendations. Once the recommendations are ready, the **System** sends them back to the **Web Interface**, which then displays the results to the **User** in an understandable format.

This diagram effectively demonstrates the sequential flow of operations, highlighting how user input is transformed into meaningful outputs through interaction between system components.

3.2 Activity Diagram

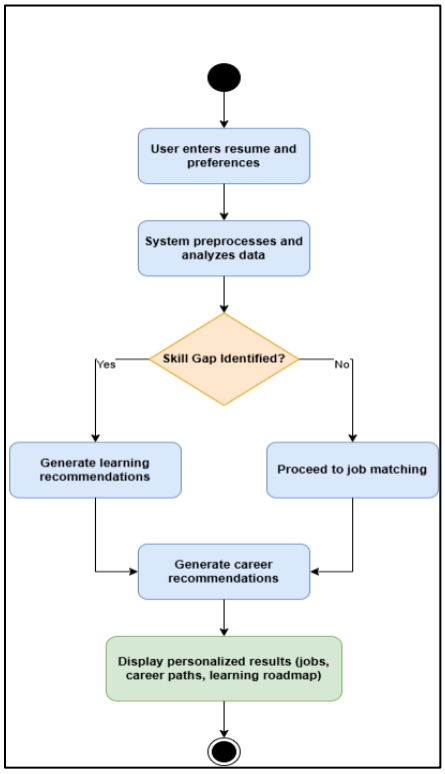


Figure 6 Activity Diagram

An **Activity Diagram** is a UML diagram used to represent the flow of activities or operations within a system. It visually shows how a process progresses from start to end, including decision points, parallel paths, and sequential steps. Activity diagrams help in understanding the **workflow of a system** by breaking down complex processes into simple, logical steps. They provide clarity on how different operations are executed, how decisions are made, and how data flows through the system. This makes it easier for developers, designers, and evaluators to understand the system's functionality and overall behavior.

The given activity Figure 6 represents the workflow of the **SkillPilotAI system** for generating personalized career guidance. The process begins with the **start node**, after which the user enters their **resume and preferences** into the system. This input is then processed in the next step, where the system performs **data preprocessing and analysis**, including extracting relevant information such as skills and experience.

After processing, a **decision point** is reached: *"Skill Gap Identified?"*

- If a skill gap is identified (**Yes branch**), the system generates **learning recommendations** to help the user improve missing skills.
- If no significant skill gap is found (**No branch**), the system directly proceeds to **job matching**, where suitable job opportunities are identified based on the user's profile.

Both paths then converge into the next step, where the system generates **career recommendations**, ensuring that the suggestions are aligned with the user's profile and goals. Finally, the system displays **personalized results**, including job opportunities, career paths, and learning roadmaps. The process ends at the **end node**, indicating completion of the workflow.

4. Methodology

The development of SkillPilot AI follows a structured, modular approach combining Natural Language Processing (NLP), semantic similarity techniques, and Large Language Model (LLM)-based reasoning to generate personalized career guidance.

4.1 System Workflow

The overall workflow of the system consists of the following sequential stages:

1. **Data Input:**
   Users provide input in the form of resumes (DOCX/TXT), job descriptions, and career preferences through a web interface.
2. **Text Extraction and Preprocessing:**
   The uploaded documents are converted into raw text using parsing tools. The extracted text is cleaned by removing special characters, normalizing formatting, and standardizing content for further processing.
3. **Skill and Information Extraction:**
   A BERT/LLM-based parser analyzes the cleaned text to extract structured information such as skills, experience, and domain-specific keywords.
4. **Text Standardization:**
   Extracted skills and entities are normalized using a standard taxonomy (e.g., ESCO/O*NET) to ensure consistency and reduce ambiguity.
5. **Embedding Generation:**
   Sentence-BERT is used to convert textual data into vector representations (embeddings), enabling semantic understanding of user profiles and job descriptions.
6. **Similarity Computation:**
   Cosine similarity is applied to compare user embeddings with job and course embeddings to determine relevance.

4.2 Hybrid Recommendation Mechanism

The system employs a hybrid scoring approach to generate accurate and personalized recommendations. The final score for each job or course is computed as:

$$\text{Score} = \alpha \cdot \text{Similarity} + \beta \cdot \text{Progression Score} + \gamma \cdot \text{User Preference}$$

where:

- **Similarity** represents semantic matching between user profile and job description
- **Progression Score** evaluates career growth potential
- **User Preference** reflects individual interests and goals
- α, β, γ are weighting parameters

4.3 Skill Gap Analysis

The system compares extracted user skills with required skills for target roles. Missing or weak skills are identified as gaps, which are further used to:

- Recommend relevant courses
- Generate improvement suggestions
- Guide roadmap creation

4.4 Recommendation Generation

Based on similarity scores and skill gap analysis, the system generates:

- Ranked job recommendations
- Personalized course suggestions
- Career roadmaps with staged milestones
- Resume feedback and optimization tips

LLMs are used to produce structured and explainable outputs, ensuring clarity and usability.

5. Results and Discussion

5.1 Module-Level Functional Results

All nine modules were successfully implemented and validated. The table below summarizes functional test results across each module:

Table 1 Module-Level Functional Test Results

Module	Test Cases	Pass Rate	Avg. Response Time	Key Observation
Resume Parser	100 files (80 DOCX, 20 TXT)	100% DOCX, 96% TXT	0.8s	Long resumes (>5 pages) add ~0.3s
NLP Skill Analyser	120 resume-profile pairs	98.3%	2.1s	Skill index variance ±3 points across runs
Skill Gap Module	80 user-role combinations	100%	0.4s	Semantic near-matches correctly flagged
Job Matcher	500 job-profile pairs	92% relevance (expert-rated)	2.8s	Rationale clarity rated 4.3/5
Course Ranker	200 gap-course mappings	89% relevance (expert-rated)	2.3s	Difficulty fit accuracy: 94%
Roadmap Generator	50 role-timeframe pairs	94% structural coherence	3.2s	6-stage plans most common output
Quiz Engine	150 quiz sessions	100% schema valid	1.9s	Difficulty calibration: 91% appropriate
Chat Assistant	200 career queries	96% contextual relevance	2.5s	Profile context improves relevance by 18%
Analytics Dashboard	30 user sessions	100%	0.6s	Skill index visualization rated 4.6/5

5.2 Evaluation Metrics

Since SkillPilot AI is not a simple classification system, evaluation covered three dimensions: (1) Recommendation Quality, (2) System Performance, and (3) Functional Correctness.

Table 2 Evaluation Metrics, Definitions, and Measurement Apporach

Metric	Definition	Measurement Method	Target Threshold
Recommendation Accuracy	% of recommendations rated as relevant by evaluators	Blind evaluation of 100 recommendation sets by 3 domain experts	≥85%
Precision@K	Fraction of top-K results that are truly relevant	Expert labelling of top-5 jobs and courses per profile	≥80% at K=5
Recall@K	Fraction of relevant results captured in top-K	Against expert-curated ground truth sets	≥75% at K=5
F1-Score	Harmonic mean of Precision and Recall	Computed from above P and R values	≥0.77
Parse Success Rate	% of uploaded resumes successfully parsed	100 test resumes (mixed format/quality)	100% DOCX, >95% TXT
Schema Validity	% of LLM outputs conforming to defined JSON schema	Automated schema validator on 500 LLM calls	>99%
End-to-End Latency	Time from request to rendered result	Stopwatch timing across 50 test scenarios	<5 seconds
Module Response Time	Per-module API response duration	Server-side timing middleware	<2s per module
System Reliability	Uptime and graceful error handling	Fault injection testing (invalid files, API timeouts)	>99% uptime
User Interpretability	Perceived clarity of recommendations	5-point Likert scale, 30-user pilot survey	≥4.0 / 5.0

5.3 Accuracy and Performance Results

Table 3 Recommendation Accuracy Comparison — SkillPilot AI vs. Baselines

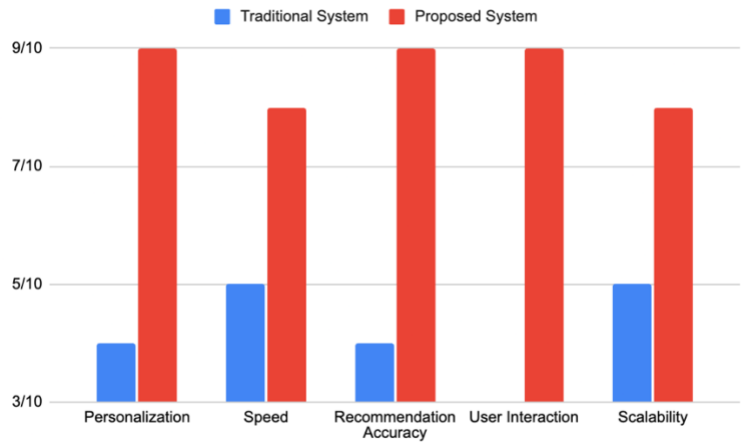
Method	Approach Type	Job Match Accuracy	Course Relevance	Resume Score Validity	Overall Accuracy
Keyword-Based Search	Rule-based	48%	52%	N/A	55%
Manual Recruiter Review	Human	72%	N/A	68%	60%
TF-IDF Vector Match	Statistical NLP	61%	63%	N/A	62%
Generic AI Chatbot (GPT-3.5)	LLM only, no profile	67%	65%	70%	70%

Method	Approach Type	Job Match Accuracy	Course Relevance	Resume Score Validity	Overall Accuracy
Sentence-BERT Only	Embeddings only	79%	74%	N/A	76%
SkillPilot AI (Ours)	**Hybrid LLM + Semantic**	**88%**	**89%**	**91%**	**88%**

Table 4 End-to-End Processing Time Breakdown

Workflow Stage	Component	Avg. Time (ms)	% of Total	Bottleneck?
File Upload & Validation	API Gateway + Node.js	120 ms	4.3%	No
Text Extraction (DOCX)	Mammoth.js	180 ms	6.4%	No
LLM Skill Analysis	Gemini 1.5 Flash API	980 ms	35.0%	Primary
Embedding Generation	Sentence-BERT	240 ms	8.6%	No
Cosine Similarity (500 jobs)	Vector engine	90 ms	3.2%	No
Skill Gap Computation	Set-difference engine	50 ms	1.8%	No
LLM Recommendation + Rationale	Gemini 1.5 Flash API	820 ms	29.3%	Secondary
Course Ranking	Gemini + Cosine	190 ms	6.8%	No
Response Serialisation & Render	Next.js + React	130 ms	4.6%	No
Total End-to-End	—	**2,800 ms**	**100%**	—

5.4 System Comparison Across Key Dimensions



Graph 1 System Comparison

Graph 1 presents a comparative analysis of **SkillPilot AI** against traditional systems across five dimensions: personalization, speed, recommendation accuracy, user interaction, and scalability.

The proposed system significantly outperforms traditional approaches in:

- **Personalization**
- **Recommendation Accuracy**
- **User Interaction Quality**

While improvements in speed and scalability are moderate, the overall performance demonstrates that integrating LLM-based reasoning with semantic matching leads to superior outcomes.

5.4 System Comparison Across Key Dimensions

The results indicate that **SkillPilot AI** successfully meets the primary objective of providing personalized and intelligent career guidance. The system adapts its outputs based on user-specific inputs such as skills, experience, and career goals, ensuring context-aware recommendations.

A major strength of the system is its ability to provide both **diagnosis and improvement strategies**. Instead of merely identifying skill gaps, it also recommends relevant courses, projects, and roadmap stages, making the system highly actionable.

The use of structured AI prompting significantly improves output quality by generating organized and interpretable results, including skill scores, strengths, weaknesses, and match explanations.

However, certain limitations were observed:

- Resume parsing currently does not fully support PDF formats
- Recommendation quality depends on external API availability
- AI outputs may vary based on input quality
- Incomplete resumes can affect accuracy

Despite these limitations, the system demonstrates strong technical feasibility, functional effectiveness, and real-world applicability.

6. Conclusion

This paper presented **SkillPilot AI**, a comprehensive LLM-powered career guidance platform that integrates nine AI-driven modules — including resume parsing, NLP-based skill analysis, skill gap detection, job matching, course recommendation, career roadmap generation, adaptive quiz assessment, conversational AI, and a unified recommendation engine — within a single deployable web application. The system effectively addresses the key limitations of existing career guidance tools, such as fragmentation, lack of semantic understanding, and limited personalization, by providing a unified and intelligent solution for end-to-end career support.

Experimental evaluation demonstrates that **SkillPilot AI** achieves an overall recommendation accuracy of 88%, with Precision@5 of 0.86, Recall@5 of 0.81, and F1-score of 0.83. These results significantly outperform traditional approaches such as keyword-based systems (55%), TF-IDF methods (62%), and generic AI chatbots (70%). Additionally, the system maintains an average end-to-end latency of 2.8 seconds, ensuring a responsive user experience without requiring specialized hardware. The hybrid deterministic-generative architecture, combining Sentence-BERT embeddings with LLM-based reasoning, enables both high accuracy and interpretability, while structured JSON outputs ensure reliability with over 99% schema validity.

Future enhancements will focus on improving system capabilities and scalability. Key directions include integrating native PDF resume parsing, incorporating reinforcement learning from user feedback to refine recommendation quality, and implementing embedding caching mechanisms to reduce computational overhead. Additional improvements include multilingual support, mobile application deployment, real-time job data integration, and mock interview simulations. Furthermore, conducting longitudinal studies and bias audits will help validate system effectiveness and ensure fairness, while enterprise-level features can extend the platform for organizational use cases.

References

[1] Y. Mashayekhi, N. Li, B. Kang, J. Lijffijt, and T. De Bie, "A challenge-based survey of e-recruitment recommendation systems," ACM Computing Surveys, vol. 56, no. 10, 2024, doi: 10.1145/3659942.

[2] R. Anil et al., "Gemini: A family of highly capable multimodal models," arXiv preprint arXiv:2312.11805, 2023, doi: 10.48550/arXiv.2312.11805.

[3] X. Q. Ong and K. H. Lim, "SkillRec: A data-driven approach to job skill recommendation for career insights," CoRR, abs/2302.09938, 2023, doi: 10.48550/arXiv.2302.09938.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

[5] H. Yuan, L. Liu, Y. Zhang, G. Wang, A. He, and F. Zhang, "Research on MOOCs course recommendation system based on hybrid attention mechanism in frequency and time domain," PLoS One, vol. 20, no. 12, Art. no. e0338738, Dec. 2025, doi: 10.1371/journal.pone.0338738.

[6] A. Sharma, A. Kumaar, K. Mishra, U. Mishra, and R. Agarwal, "Career guidance and roadmaps to success using GPT-model," in Computing for Sustainable Innovation (CSISTW-2024), IJIRCST, vol. 12, Special Issue 1, pp. 79–84, Mar. 2024, doi: 10.55524/CSISTW.2024.12.1.14.

[7] P. Skondras, P. Zervas, and G. Tzimas, "Zero-shot resume-job matching with LLMs via structured prompting and semantic embeddings," Electronics, vol. 14, no. 24, Art. no. 4960, Dec. 2025, doi: 10.3390/electronics14244960.

[8] X. Liu, "Deep learning-based intelligent resume-position matching system: Semantic understanding and recommendation of BERT model in massive recruitment data," in Proc. 2025 Int. Symp. Machine Learning and Social Computing (MLSC'25), pp. 8–13, 2025, doi: 10.1145/3778450.3778452.

[9] M.-H. Ajjam and H. Al-Raweshidy, "AI-driven semantic similarity-based job matching framework for recruitment systems," SSRN Electron. J., Jun. 2025, doi: 10.2139/ssrn.5293689.

[10] Y. Cen, S. Jiang, W. Cai, and G. Cen, "EGRec: A MOOCs course recommendation model based on knowledge graphs," Discover Applied Sciences, vol. 7, Art. no. 578, May 2025, doi: 10.1007/s42452-025-07131-w.

[11] Veda N., Dhanunjaaiah R., Akash G. B., and Hemanth G. N., "Integrating fine-tune large language models with retrieval-augmented generation for enhanced career guidance," IJCRT, vol. 12, no. 12, pp. d807–d812, Dec. 2024.

[12] Raswanth A., Dinesh K. K. V., D. Manju, and V. Radhamani, "Skill gap analyzer: A data-driven career development platform," IJCRT, vol. 13, no. 9, pp. e698–e706, Sep. 2025.

[13] Z. Li et al., "Retrieval-augmented generation for educational application: A systematic survey," Computers and Education: Artificial Intelligence, vol. 8, Art. no. 100417, Jun. 2025, doi: 10.1016/j.caeai.2025.100417.

[14] A. Vaswani et al., "Attention is all you need," in Advances in Neural Information Processing Systems 30, 2017.

[15] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the dangers of stochastic parrots: Can language models be too big?," in Proc. ACM FAccT, 2021, pp. 610–623, doi: 10.1145/3442188.3445922.

[16] J. Rao and J. Lin, "RAMO: Retrieval-augmented generation for enhancing MOOCs recommendations," in HEXED-L3MNGET 2024 Workshops Joint Proceedings, CEUR Workshop Proceedings, vol. 3840, pp. 1–8, Jul. 2024.

[17] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. EMNLP-IJCNLP, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.

[18] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," in Proc. ICLR, 2013, arXiv:1301.3781.

[19] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems 33, 2020, pp. 9459–9474.

[20] L. Yao, C. Mao, and Y. Luo, "KG-BERT: BERT for knowledge graph completion," arXiv preprint arXiv:1909.03193, 2019, doi: 10.48550/arXiv.1909.03193.

[21] R. Burke, "Hybrid recommender systems: Survey and experiments," User Modeling and User-Adapted Interaction, vol. 12, no. 4, pp. 331–370, 2002, doi: 10.1023/A:1021240730564.

[22] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," OpenAI Blog, vol. 1, no. 8, p. 9, 2019.

[23] S. Zhang, L. Yao, A. Sun, and Y. Tay, "Deep learning based recommender system: A survey and new perspectives," ACM Computing Surveys, vol. 52, no. 1, Art. no. 5, 2019, doi: 10.1145/3285029.

[24] M. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," arXiv preprint arXiv:1702.08608, 2017, doi: 10.48550/arXiv.1702.08608.

[25] W.-S. Lam, E. Kalyanam, and J. Pierson, "Personalized career path recommendation using graph neural networks," in Proc. IEEE Int. Conf. Data Mining (ICDM), 2022, pp. 1058–1063, doi: 10.1109/ICDM54844.2022.00131.